

Children's Color Acquisition Across Cultural Environments in the Age of Algorithms

Version 1.0.0

An Initial Conceptual Research Preprint

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Abstract

The research examines how algorithmic image culture relates to culturally determined color acquisition in childhood. The recognition, discrimination, naming, and categorization of colors is not exclusively a perceptual and developmental process, but also an experience shaped by the social and cultural environment. The child encounters colors through objects, images, linguistic naming, visual conventions, and cultural symbols, meaning that the meanings and uses associated with colors may be culturally mediated from the early stages of development.

The spread of digital image culture places this visual experience under new conditions. With the emergence of generative artificial intelligence, images and their visual properties can be created, modified, and reconfigured within a short period of time. In this environment, color does not necessarily appear as a stable property of an object or image, but may also emerge as a visual element that can be altered and recreated through different instructions and algorithmic processes.

The central question of the research is therefore not how artificial intelligence in itself changes children's color perception, but rather how the algorithmically mediated visual environment connects to already existing cultural patterns during color acquisition. Particular attention is given to how algorithmic image generation may encounter different visual traditions, color associations, social meanings, and everyday practices.

The research starts from the premise that the same algorithmic image-making systems may appear as part of different visual experiences in different cultural environments. The global nature of the technological system therefore does not necessarily result in a uniform experience of color or a uniform visual learning environment. Algorithmic image culture and the cultural environment may interact in shaping which colors, combinations, visual patterns, and meanings become familiar.

The aim of the study is to develop a theoretical framework that interprets children's color acquisition in perceptual, developmental, cultural, and technological contexts. The present V1 preprint does not aim to present empirical results, but rather to conceptually define the problem and establish the theoretical foundations necessary for subsequent cross-cultural empirical investigations.

Keywords

color acquisition, color perception, childhood development, algorithmic image culture, generative artificial intelligence, digital image culture, visual experience, cultural context, cross-cultural research, visual culture

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Introduction

The development of children's recognition and acquisition of colors is a gradual developmental process. Over time, young children learn to distinguish, recognize, name, and categorize different visual characteristics, to which they later also associate different meanings. This development cannot be attributed exclusively to the biological and perceptual characteristics of perception, as experiences with colors are also shaped by encounters with objects, images, other people, and cultural symbols.

The meanings, uses, and visual conventions associated with color may differ across societies. The same chromatic phenomenon may carry different emotional, religious, aesthetic, or everyday interpretations, meaning that acquisition always takes place within a specific socio-cultural context. This context influences not only which shades and color combinations become familiar, but also what meanings and associations become connected to them.

The emergence of digital image culture has supplemented this learning environment with new visual experiences. Through content displayed on screens and digital image-making systems, children increasingly encounter images whose creation and appearance are independent of the immediate physical environment. Generative artificial intelligence introduces a further characteristic within this development: an image or individual elements of it can be recreated and modified within a short period of time. An object or scene can thus appear with different colors, shades, and visual styles without the physical referent itself changing.

This possibility changes the conditions under which color appears visually. The chromatic property is not necessarily tied to a particular object or a single stable representation, but may also appear as a variable visual parameter. This alone, however, does not imply a change in color perception or color acquisition. The more relevant question is how such image-making practices connect to already existing cultural and visual experiences.

This connection is particularly important from a cross-cultural perspective. Although the technological possibilities of algorithmic image-making systems may be globally similar, their

use and the content they produce may be shaped by local factors. Family and educational practices, visual traditions, social norms, and digital habits may all play a role in determining which images become part of a child's everyday experience. The same technological system may therefore become associated with different visual references and color associations in different cultural contexts.

The present research starts from this connection. Its central question is how algorithmic image culture relates to culturally determined color acquisition in childhood. The study does not assume that artificial intelligence independently and uniformly transforms the developmental process, but rather seeks to examine how algorithmically generated images relate to already existing cultural patterns and what role this relationship may play in the development of the child's visual experience.

The present V1 preprint provides a theoretical foundation for this problem. Its aim is to develop a conceptual framework that interprets color acquisition in perceptual, developmental, cultural, and technological contexts, and that can provide a basis for subsequent cross-cultural empirical investigations. The study therefore defines the fundamental concepts and main points of connection of the investigation, and identifies the comparative dimensions along which processes taking place in different cultural environments can later be examined.

1. Color Acquisition as a Developmental Process

Encountering color is present from the early stages of a child's development; however, the acquisition of color does not refer to the development of a single ability. The child gradually learns to perceive visual differences, distinguish, recognize, name, and categorize colors. Throughout this process, color increasingly becomes a conceptual unit to which the child may associate stable labels and, later, social, emotional, or cultural meanings.

Color perception and color acquisition are therefore not identical concepts. The perceptual system enables the differentiation of light stimuli of different wavelengths, but this does not automatically mean that a child associates a particular color with the same category, name, or meaning as another child. During development, perceptual experience gradually becomes connected to language, categorization, and the rules of use learned from the environment.

Color acquisition thus encompasses several interconnected levels. The first level is the perception of differences, followed by the recognition and differentiation of colors. This is followed by linguistic naming and then by the organization of colors into categories. During later development, social and cultural meanings associated with color may also become incorporated into the child's experience.

It follows that color acquisition is not simply the unfolding of a biological predisposition. Biological and perceptual foundations are necessary, but they do not, by themselves, explain which color categories a child learns, which names they use, which objects they associate with a particular color, or what meaning they attribute to it.

Within the present research framework, color acquisition is therefore understood as a developmental process in which perceptual experience gradually becomes connected to the linguistic, social, and cultural environment. This definition makes it possible to examine color acquisition not exclusively as an individual developmental process, but also as an experience that is dependent on the environment.

2. Color in a Cultural Context

A child's experience with color does not arise in isolation, but within a cultural space. Colors appear in the objects of everyday life, clothing, images, toys, built environments, festive events, and various social practices. These repeated encounters may contribute to the child acquiring colors not merely as visual properties, but as elements associated with certain objects, situations, and meanings.

In this sense, the cultural environment is not simply a background to development. It may determine which visual patterns the child encounters regularly, which colors are emphasized, and what social meanings are associated with them. A particular color may appear, for example, in festive, religious, social, or everyday contexts, and these recurring associations may shape the child's later interpretations.

Cultural difference does not necessarily mean that people perceive colors in fundamentally different ways. Distinguishing between the foundations of perceptual system functioning and culturally developed meanings is therefore of fundamental importance. The same visual color may be associated with different objects, situations, or symbolic meanings in different societies.

The child therefore does not merely acquire a color name, but learns to use and interpret color within an environment in which its social meaning also develops. In this process, color may become connected to language, material culture, social norms, and visual traditions.

This cultural embeddedness makes it possible to examine color acquisition across cultures. If the child's experience with color is partly shaped by the cultural environment surrounding them, then different developmental experiences and systems of meaning may emerge in different societies.

3. The Child's Digital Visual Environment

Children's visual experience has gradually expanded into digital environments over the past decades. Images displayed on screens, digital games, videos, animations, and online content have

become part of everyday visual experience. This has not eliminated the role of the physical environment, but has supplemented it with a new visual layer.

An important characteristic of the digital image is that children may also encounter visual content that does not originate directly from the physical environment. A digital image can depict objects, places, and color combinations that the child does not encounter in their own surroundings. The digital visual environment can therefore expand the child's experiential space.

At the same time, digital images are not culturally neutral content. Digital visual content created for children is produced and used by different societies, media systems, and cultural practices. As a result, the digital environment remains connected to the society in which the child lives.

From this perspective, generative artificial intelligence does not represent the beginning of digital visual culture, but rather a new stage of it. In the earlier digital environment, children mostly encountered images that had already been created. In the case of generative systems, however, the creation and modification of images also become easier.

This difference leads to the question of what color acquisition means within a visual environment in which images can not only be viewed, but can also be increasingly easily created and transformed.

4. Cultural Application of AI

The global presence of algorithmic technology does not mean that its use takes place in the same way in every society. The possibilities provided by the technology may be similar, but the content created by people and the purposes for which the technology is used may be shaped by cultural, social, and economic circumstances.

Generative AI therefore does not determine the visual environment on its own. Users choose what kinds of images they create, which objects and characters they depict, which colors they use, and for what purposes they apply the generated content. These decisions may be connected to local cultural norms and visual traditions.

The same algorithmic system may therefore lead to different visual outcomes in different cultural environments. The technology may be global, while its specific visual application remains local. For a child, AI does not appear as an abstract technological system, but through the images, stories, games, and visual content encountered in their everyday environment.

Based on this, cultural application may become one of the determining mediating layers of the algorithmic visual environment. It is therefore not sufficient to establish whether artificial intelligence is present in a country. From the perspective of the research, it is also necessary to examine how the technology enters children's visual environments.

Cultural difference is thus not necessarily a difference in the presence or absence of AI, but may instead be a difference in how AI is used and culturally embedded.

5. Possible Cultural and Social Consequences

The emergence of algorithmic image culture in itself does not make it possible to state that children's color acquisition will necessarily change. The present research framework therefore does not assume a predetermined effect, but distinguishes several possible directions.

The first possibility is cultural continuity. In this case, generative AI is used to create visual content that carries forward the existing color culture of a given society. In this case, the algorithmic tool provides a new technological form for already existing cultural patterns.

The second possibility is cultural transformation. In this case, algorithmically generated images create new combinations and new visual relationships that may, over time, modify previously familiar color–object or color–meaning associations.

The third possibility is cultural homogenization. Globally used algorithmic systems and widely repeated visual patterns may make certain color palettes more similar across different societies. In this case, cultural differences may decrease in certain areas.

The fourth possibility is the opposite: cultural diversification. AI may provide different societies with a tool that enables the easier reproduction and transmission of local visual traditions, colors, and cultural symbols.

These four possibilities should not be understood as mutually exclusive outcomes, but as directions for research. In some cultural environments, continuity may be stronger, while in others, transformation, homogenization, or diversification may predominate. The question therefore requires empirical investigation.

At the V1 stage, the aim is not to identify any of these possibilities as a demonstrated outcome, but to determine through which processes they may later become measurable.

6. Research Questions

Based on the theoretical framework outlined above, the central question of the research is how children's color acquisition develops in different cultural environments when generative artificial intelligence becomes part of the visual environment.

The first question aims to explore what cultural differences can be observed in children's everyday experiences with color and in their systems of meaning.

The second question examines how different societies incorporate visual content created by generative AI into children's environments.

The third question addresses how colors that can be algorithmically created and modified relate to children's already developing color categories, color names, and meanings associated with color.

The fourth question examines how the same algorithmic technology may result in different visual experiences in different cultural environments.

The fifth question addresses whether algorithmic image culture, in the longer term, is more likely to maintain, transform, or reduce cultural differences among children in relation to color.

These questions build upon one another. The first establishes the cultural starting point, the second examines the cultural integration of AI, the third focuses on the process occurring at the level of the child, the fourth addresses differences between cultures, and the fifth considers possible longer-term social consequences.

In their current form, these questions are not empirical hypotheses, but guiding questions for a subsequent research program.

7. Directions for Future Empirical Research

The present V1 preprint provides a theoretical starting point; therefore, the task of the next stage of research is to make the concepts defined here empirically testable. To this end, comparable cultural environments must first be identified in which children's visual and social environments can be adequately described.

The comparison should not be based solely on country names. It should examine local visual culture, the visual content consumed by children, linguistic color categories, the social meanings of colors, as well as the forms in which generative AI appears in everyday life.

The second level of research could involve examining children's actual experiences with color. This may include the investigation of color recognition, color naming, categorization, color-object associations, and cultural meanings associated with colors.

The third level could involve analyzing the algorithmically created visual environment. Here, it is necessary to examine what kinds of images and color palettes are created or used with AI systems in different cultural environments, particularly in the case of content produced for children.

The fourth level is the cultural comparison. The same or functionally equivalent tasks could be applied in different countries and cultural environments to examine how algorithmic visual experience manifests itself with different characteristics.

At the same time, future research must distinguish biological and perceptual characteristics from cultural and technological influences. Only in this way can it be determined whether an observed difference is actually related to the cultural environment or to algorithmically mediated visual experience.

The ultimate goal of the research program is not to prove that AI uniformly changes children's color acquisition. The aim is to explore how color acquisition, the cultural environment, and algorithmically mediated visual experience are related to one another across different societies.

This V1 should therefore not be understood as a completed research result, but as the starting point for a subsequent cross-cultural empirical research program.

Conclusion

Children's color acquisition is a developmental process in which the cultural environment may play a determining role alongside perceptual abilities. Children do not encounter colors in isolation, but acquire their recognition, naming, categorization, and meanings within the visual, linguistic, and cultural world of a given society.

The emergence of generative artificial intelligence creates new conditions within this process. The possibility of rapidly creating and modifying colors and visual elements changes the nature of the child's visual environment. At the same time, this change is not independent of culture. AI technologies are applied differently across different societies and may therefore create different cultural visual environments.

The present V1 preprint does not seek to provide an answer as to exactly how these changes affect children's color acquisition. Its aim is to define the problem and establish a theoretical starting point that makes possible the subsequent cross-cultural and empirical investigation of the question.

In this sense, the V1 preprint represents a starting point for the research: a first step toward understanding what new developmental and social questions arise from the intersection of color acquisition, the cultural environment, and generative artificial intelligence.

Research Ethos Statement

This research is presented as an independent theoretical and conceptual inquiry. Version 1.0 does not present an empirical study or claim empirically established effects of color on children's perception or learning. No empirical data were collected or analyzed in this version. The propositions presented here are intended as a preliminary conceptual framework for subsequent investigation.

The research distinguishes between observations, conceptual propositions, and findings that would require empirical verification. Accordingly, the present version does not assume that particular colors produce specific perceptual, psychological, or cultural effects, nor does it treat the possibilities of generative artificial intelligence as established empirical findings.

The purpose of Version 1.0 is to establish the research problem and its initial conceptual boundaries while leaving its empirical development open. Subsequent research may confirm, refine, qualify, or reject the propositions introduced in this version.

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